

An Efficient Dynamic Workload Balancing Strategy

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[Abstract]

In this paper, we propose an efficient dynamic workload balancing strategy which improves the performance of high-performance computing system. The key idea of this dynamic workload balancing strategy is to minimize execution time of each job and to maximize the system throughput by effectively using system resource such as CPU, memory. Also, this strategy dynamically allocates job by considering demanded memory size of executing job and workload status of each node. If an overload node occurs due to allocated job, the proposed scheme migrates job, executing in overload nodes, to another free nodes and reduces the waiting time and execution time of job by balancing workload of each node. Through simulation, we show that the proposed dynamic workload balancing strategy based on CPU, memory improves the performance of high-performance computing system compared to previous strategies.

▶ **Key words** : Allocation, Workload, Migration, Load balancing, Simulation

[요 약]

논문에 대한 요약은 기술하시면 됩니다. 요약을 작성하실 때에는 문단을 구분하지 마시고, 논문의 연구목적, 연구방법, 분석결과, 결론을 제시하여 주시기 바랍니다.

▶ **주제어** : 할당량, 작업량, 이송, 부하균형, 시뮬레이션

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I. Introduction

한국컴퓨터정보학회 발전 방안을 위한 서론을 기술하고 있으며, 본 방안을 수행하기 위한 본 학회의 발전은 다음과 같이 기술할 수 있으며, 향후 논의를 통해 발전 방안을 제시할 수 있습니다[1][3-5]. 추가적으로 기록 할 내용은 연구내용과 함께

II. Preliminaries

1. Related works

1.1 국내 동향

한국컴퓨터정보학회 발전방안을 위한 관련연구로서 다양한 의견들이 제시되고 있다. 먼저, 발전방안의 기술을 살펴보면 이해그룹의 장점과 단점을 살펴봄, 이를 통하여..

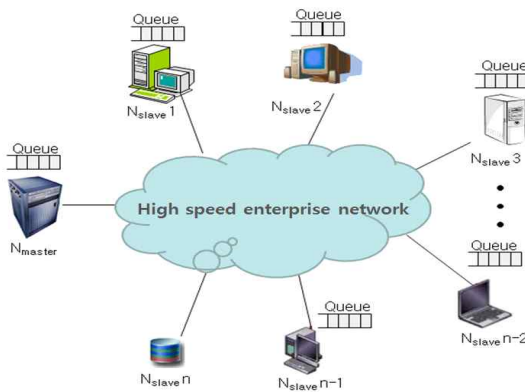


Fig. 1. System Architecture

먼저, 발전방안의 기술을 살펴보면 이해 그룹의 장점과 단점을 살펴봄,

Table 1. System Environment

Item	Value
CPU Clock Speed	100 ~ 500 MIPS
Memory Size	32 ~ 256 MB
System File Size	16 MB

III. The Proposed Scheme

한국컴퓨터정보학회 발전방안을 위한 관련연구로서 다양한 의견들이 제시되고 있다. 먼저, 발전방안의 기술을 살펴보면 이해그룹의 장점과 단점을 살펴봄, 이를 통하여..

IV. Conclusions

한국컴퓨터정보학회 발전방안을 위한 관련연구로서 다양한 의견들이 제시되고 있다. 먼저, 발전방안의 기술을 살펴보면 이해그룹의 장점과 단점을 살펴봄, 이를 통하여..

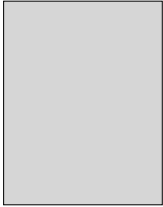
ACKNOWLEDGEMENT

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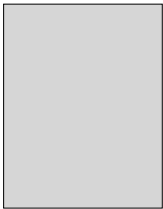
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